**10 In this question you must show detailed reasoning.**

The equation of a curve is

$$y = \frac{\sin 2x - x}{x \sin x}.$$

**(a)** Use the small angle approximation given in the list of formulae on pages 2–3 of this question paper to show that

$$\int_{0.01}^{0.05} y \, dx \approx \ln 5. \quad \textbf{[4]}$$

**(b)** Use the same small angle approximation to show that

$$\frac{dy}{dx} \approx -10000 \text{ at the point where } x = 0.01. \quad \textbf{[2]}$$

The equation $y = 0$ has a root near $x = 1$. Joan uses the Newton-Raphson method to find this root. The output from the spreadsheet she uses is shown in Fig. 10.1.

| $n$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| $x_n$ | 1 | 0.958509 | 0.950084 | 0.948261 | 0.94786 | 0.947772 | 0.947753 | 0.947748 |

**Fig. 10.1**

Joan carries out some analysis of this output. The results are shown in Fig. 10.2.

| $x$ | $y$ |
|---|---|
| 0.9477475 | –7.79967E–07 |
| 0.9477485 | –2.90821E–06 |
| $x$ | $y$ |
| 0.947745 | 4.54066E–06 |
| 0.947755 | –1.67417E–05 |

**Fig. 10.2**

**(c)** Consider the information in Fig. 10.1 and Fig. 10.2.

- Write 4.54066E–06 in standard mathematical notation.
- State the value of the root as accurately as you can, justifying your answer. **[3]**

**10(a)**

**10(b)**

| | |
|---|---|
| **10(c)** | |
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